

2018A F=ma Exam: Problem 11

Kevin S. Huang

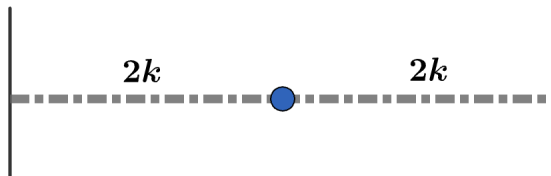
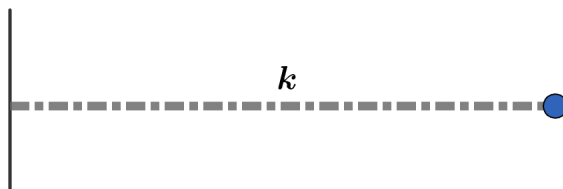
Recall the angular frequency of a mass-spring system is given by

$$\omega = \sqrt{\frac{k}{m}}$$

A spring consists of two ‘half-springs’ in series. The reciprocal of the effective spring constant of two springs in series is the sum of their reciprocals. Thus,

$$\frac{1}{k} = \frac{1}{k_{1/2}} + \frac{1}{k_{1/2}}$$

so the spring constant of a ‘half-spring’ is $k_{1/2} = 2k$.



The effective spring constant of two springs in parallel is their sum. Therefore,

$$k' = k_{1/2} + k_{1/2} = 4k$$

$$\omega' = \sqrt{\frac{4k}{m}} = 2\omega$$

so the answer is D.